Optimized Hybrid Grey–Fourier Modeling for Precision Landslide Displacement Forecasting: A Comparative Study of Classical and Metabolic GM(1,1)

Dhakshin N, Rohith Ravi, Sanjay Kumar S, and Akhilan

***Abstract*—Every year, landslides claim thousands of lives and cause property damage that dwarfs the resources available for recovery. What makes them particularly brutal is the timing—by the time movement becomes visible, the window for meaningful intervention has often already closed. Getting displacement forecasts right, then, is not an academic exercise; it is the foundation any credible early-warning system has to be built on.**

This paper presents a Hybrid Grey–Fourier framework that pairs two versions of the GM(1,1) grey model with an FFT residual-correction step. The Classical version fits one GM(1,1) model across the full monitoring history. The Metabolic version works differently: it keeps only the $L = 5$ most recent readings, discards older ones, and re-estimates its parameters each time a new observation comes in. Both pass their residuals through an FFT layer that identifies and corrects the periodic errors a grey model on its own will never clean up.

We tested the framework on the Preonzo slope dataset (2002–2012), which covers five extensometer sensors (E1–E5) on a Swiss hillside that eventually gave way on 15 May 2012. A composite risk index— combining relative prediction error with the instantaneous development coefficient a—flags each time step as either Safe or Advanced Warning. In the days just before failure, the Metabolic model with Fourier correction pulled well ahead of the Classical variant: RMSE dropped to 0.7230 mm and MAE to 0.4449 mm, compared to 5.0549 mm and 3.5191 mm on sensor E1. More critically, the global risk index tripped the warning threshold roughly 2–3 days before the slope collapsed—enough lead time to matter if an evacuation plan is already in place.

***Index Terms*—Grey model, GM(1,1), sliding window, metabolic grey model, Fast Fourier Transform, landslide prediction, displacement forecasting, risk index, Preonzo slope.**

I. Introduction

LANDSLIDES sit in an uncomfortable category of disasters—not rare enough to ignore, yet not predictable enough to manage reliably. The gap between first detectable movement and actual failure is often measured in hours or days, and without a warning system that can actually keep pace with that timeline, even a moderate-sized slide can produce casualties far out of proportion to what any reasonable mitigation effort would suggest. Global fatality counts have remained stubbornly high year after year—a fairly clear signal that existing approaches, whatever their individual merits, are still leaving gaps [1].

Landslide prediction methods generally split into two camps. Mechanical models build from the physics: stress redistribution, frictional resistance, pore-pressure buildup. Their outputs are interpretable in geomechanical terms, which is genuinely useful. The catch is that they require detailed site characterisation—the kind that takes time and money to acquire and is often simply unavailable at remote locations.

Mathematical models sidestep this entirely by working directly from monitoring records, without attempting to represent the underlying physics explicitly. It has been recognised for some time that displacement on an unstable slope tends to follow a recognisable three-stage pattern, illustrated schematically in Fig. 1: an initial deceleration phase (AB), a period of roughly linear steady creep (BC), and then a runaway acceleration (CD) that signals imminent failure.

Within the mathematical toolkit, grey models occupy a specific niche. Deng developed them precisely for settings where data are scarce and noisy [2]—which describes field instrumentation on unstable slopes fairly accurately. The standard GM(1,1) has two practical limitations, though. First, it is restricted to a monotonic exponential trajectory, meaning any seasonal or cyclical displacement pattern will produce structured residuals the model never resolves on its own. Second, fitting a single exponential to the full history forces the parameter estimates to absorb information from months or years back that may no longer reflect what the slope is currently doing.

Both problems are addressed in the present work. We couple GM(1,1) with a sliding-window mechanism—the Metabolic variant—that limits parameter estimation to the L most recent observations at each step, and attach an FFT correction layer that removes periodic error components before each prediction is finalised.

The remainder of the paper is organised as follows. Section II reviews relevant prior work. Section III describes the methodology in detail. Section IV reports experiments on the Preonzo dataset. Section V discusses the findings, and Section VI concludes.

II. Related Works

Grey System Theory was introduced by Deng in 1982 [2], originally motivated by control problems in systems where

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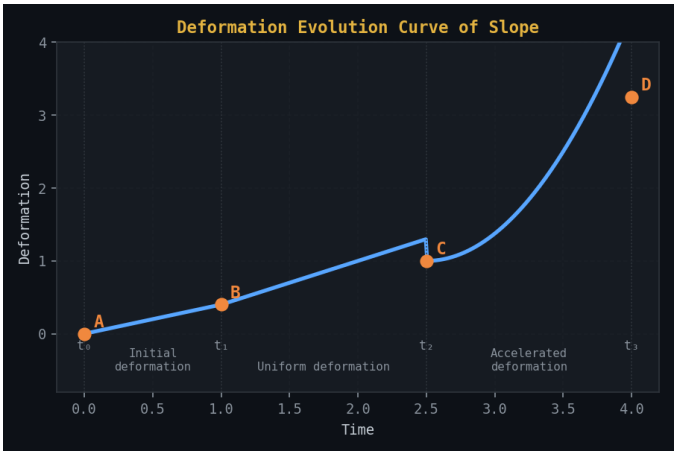

Fig. 1. Deformation evolution curve of slope.

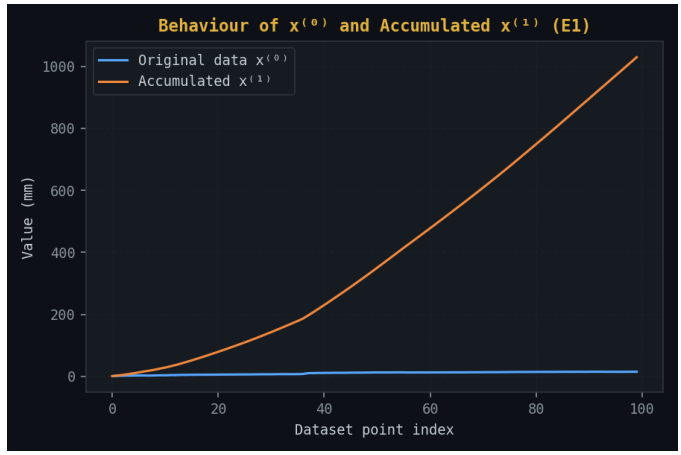

Fig. 2. Behaviour of original data $x^{(0)}$ and accumulated sequence $x^{(1)}$ (E1 sensor, first 100 data points).

only partial information is available. The observation that useful predictions can be extracted from small, irregular data sequences led to adoption well beyond the original domain. Applications have since accumulated in population dynamics [3], crop yield estimation [4], macroeconomic forecasting [5], and energy-consumption modelling [6], among others.

The base GM(1,1) is inherently limited in its forecast horizon, and this has driven a substantial body of extension work. Zhang et al. introduced nonlinear time-correction factors, producing the TGM(1,1,p,q) family [7]. Li and Wu took a different approach, pairing grey modelling with Twin Support Vector Regression and a Hausdorff Derivative Operator to handle stronger nonlinearities [8]. Wu et al. added a quadratic time term in their BNGM(1,1,t^2) to better capture accelerating trends [9]. Li et al. explored fractional-order calculus as a unifying framework for grey models with memory-like behaviour [10]. For life-expectancy estimation, Liu et al. found that combining grey models with machine-learning components improved generalisation considerably [11].

Non-grey approaches are of course widely represented in the geohazard literature. Artificial neural networks [12], [13], support vector machines [14], genetic-algorithm-guided neural networks [15], decision trees [16], and random forests [17] all appear regularly. PSO–SVM coupling has shown particular promise for slope-specific problems [18], and more recently deep architectures combining sparse autoencoders with metaheuristic search have been validated on large monitoring datasets [19].

What sets the present approach apart is a deliberate preference for simplicity. The pipeline consumes only univariate displacement data, runs fast enough for field deployment, and requires no deep-learning infrastructure to maintain or retrain.

III. Methodology

The framework has four components: (A) Classical GM(1,1) with Fourier correction, (B) Metabolic GM(1,1) with the same correction, (C) accuracy metrics, and (D) the risk index and warning logic.

A. Data Preprocessing

We denote the raw displacement sequence as $\mathbf{x}^{(0)} = \{x^{(0)}(1), x^{(0)}(2), \ldots, x^{(0)}(n)\}$. GM(1,1) requires strictly positive input, so we apply the offset transformation:

$$x^{(0)}(k) \leftarrow \left|x^{(0)}(k)\right| - \min_j \left|x^{(0)}(j)\right| + 1. \tag{1}$$

The additive offset of 1 ensures the minimum mapped value stays positive even when the raw minimum is zero.

B. Classical GM(1,1) + Fourier

1) Accumulated Generating Operation (AGO): Fitting a differential equation directly to noisy displacement increments is numerically ill-conditioned. The 1-AGO transformation works around this by operating on the running sum:

$$x^{(1)}(k) = \sum_{i=1}^{k} x^{(0)}(i), \quad k = 1, 2, \ldots, n. \tag{2}$$

Fig. 2 shows the resulting accumulated sequence against the original data for the first 100 readings of the E1 sensor.

2) Background Value Sequence: The background value sequence is a weighted average of successive AGO values:

$$z^{(1)}(k) = \lambda\, x^{(1)}(k) + (1-\lambda)\, x^{(1)}(k-1), \quad k = 2, \ldots, n, \tag{3}$$

where $\lambda \in (0, 1)$ controls the interpolation point. The standard choice $\lambda = 0.5$ amounts to the trapezoidal rule and works well for roughly linear growth; values below 0.5 suit rapidly accelerating sequences; $\lambda \to 1$ is more appropriate when a sequence is approaching a plateau. We use $\lambda = 0.6$ throughout, which performed consistently across all five sensors in preliminary trials.

3) Regularised Parameter Estimation: The core grey relation between the raw and accumulated sequences is:

$$x^{(0)}(k) + a\, z^{(1)}(k) = b, \quad k = 2, \ldots, n. \tag{4}$$

Writing this for each time step gives the overdetermined system $\mathbf{Y} = \mathbf{B}\boldsymbol{\theta}$:

$$\mathbf{B} = \begin{bmatrix} -z^{(1)}(2) & 1 \\ -z^{(1)}(3) & 1 \\ \vdots & \vdots \\ -z^{(1)}(n) & 1 \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} x^{(0)}(2) \\ x^{(0)}(3) \\ \vdots \\ x^{(0)}(n) \end{bmatrix}. \tag{5}$$

Ordinary least squares becomes numerically unstable when $\mathbf{B}^\top\mathbf{B}$ is near-singular—a situation that arises regularly over short windows. We add a Ridge penalty $\lambda_r\|\boldsymbol{\theta}\|^2$ to stabilise the solution:

$$\hat{\boldsymbol{\theta}} = [a,\, b]^\top = \left(\mathbf{B}^\top\mathbf{B} + \lambda_r\mathbf{I}\right)^{-1}\mathbf{B}^\top\mathbf{Y}, \quad \lambda_r = 0.1. \tag{6}$$

This follows from minimising $J = \|\mathbf{Y} - \mathbf{B}\boldsymbol{\theta}\|^2 + \lambda_r\|\boldsymbol{\theta}\|^2$.

The two fitted parameters carry physical meaning worth keeping in mind. The development coefficient a captures how rapidly the process is evolving: a small negative a signals slow, steady creep, while a large $|a|$ points to accelerating instability. The grey driving quantity b represents a roughly constant background forcing, such as the sustained pore-pressure contribution from seasonal rainfall.

4) Whitenisation and Time-Response Function: The continuous analogue of (4) is the first-order ODE:

$$\frac{dx^{(1)}(t)}{dt} + a\,x^{(1)}(t) = b. \tag{7}$$

Multiplying through by e^{at} and applying the initial condition $x^{(1)}(1) = x^{(0)}(1)$ yields the time-response function:

$$\hat{x}^{(1)}(k) = \left(x^{(0)}(1) - \frac{b}{a}\right)e^{-a(k-1)} + \frac{b}{a}. \tag{8}$$

Original-scale predictions are recovered by the inverse AGO:

$$\begin{aligned} &\hat{x}^{(0)}(1) = x^{(0)}(1), \\ &\hat{x}^{(0)}(k) = \hat{x}^{(1)}(k) - \hat{x}^{(1)}(k-1), \quad k = 2, \ldots, n. \end{aligned} \tag{9}$$

5) FFT Residual Correction: After the inverse AGO step, the residuals $e(k) = x^{(0)}(k) - \hat{x}^{(0)}(k)$ are nowhere near white noise on the Preonzo dataset. They carry clear periodic structure tied to hydrological forcing—annual snowmelt, semi-annual rainfall peaks, and shorter weather-driven pulses. Modelling these periodicities inside the grey framework is inefficient; the Discrete Fourier Transform handles them far more naturally:

$$E(m) = \sum_{k=1}^{n} e(k)\exp\left(-j\frac{2\pi(m-1)(k-1)}{n}\right). \tag{10}$$

In practice we use the FFT algorithm (complexity $O(n\log n)$). We retain the $K = 3$ dominant positive-frequency components—capturing roughly the annual, semi-annual, and short-term rainfall periodicities—together with the DC offset, enforce conjugate symmetry, and reconstruct the filtered residual by the Inverse DFT:

$$\hat{e}(k) = \frac{1}{n}\sum_{m=1}^{n} \tilde{E}(m)\exp\left(j\frac{2\pi(m-1)(k-1)}{n}\right). \tag{11}$$

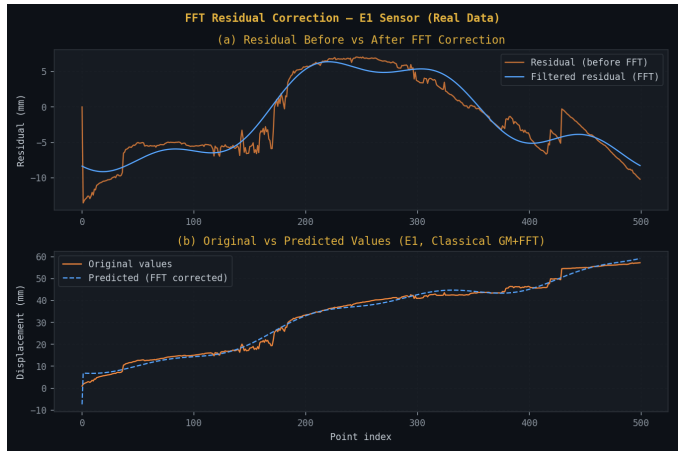

Fig. 3. FFT residual correction on the E1 sensor. (a) Residual before and after FFT filtering. (b) Original vs. predicted values after correction.

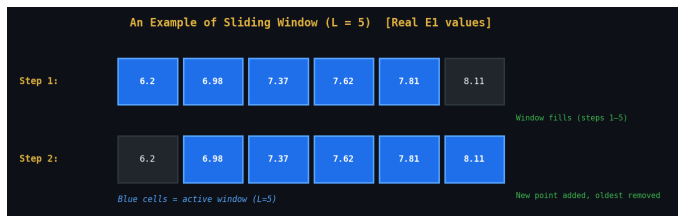

Fig. 4. An example of sliding window ($L = 5$).

Adding this correction back to the grey prediction gives the final output:

$$\hat{x}^{(0)}_{\text{final}}(k) = \hat{x}^{(0)}(k) + \hat{e}(k). \tag{12}$$

Fig. 3 shows the correction applied to the E1 sensor.

C. Metabolic GM(1,1) + Fourier

1) Sliding Window Definition: The most serious limitation of the classical variant for pre-failure forecasting is that data from three years ago receives the same weight as data from yesterday. The Metabolic variant fixes this by estimating parameters from only the L most recent observations at each step t:

$$\mathbf{x}^{(0)}_w(t) = \left\{x^{(0)}(t-L), \ldots, x^{(0)}(t-1)\right\}, \quad L = 5. \tag{13}$$

Fig. 4 illustrates the windowing scheme.

2) Window Prediction: At each step a fresh ridge estimator—identical to (6) but applied to L rows only—produces window-specific parameters (a_t, b_t). The one-step-ahead prediction is then:

$$\hat{x}^{(0)}_{\text{meta}}(t) = \hat{x}^{(1)}_w(L) - \hat{x}^{(1)}_w(L-1), \tag{14}$$

where $\hat{x}^{(1)}_w(\cdot)$ is computed from the window-specific parameters. The sequence $\{a_t\}$ encodes instantaneous slope acceleration and feeds directly into the risk index below. FFT correction is applied to the metabolic residuals in exactly the same way as (10)–(12).

Algorithm 1 Hybrid Grey–Fourier Prediction and Warning

Require: Sequence $\{x^{(0)}(k)\}$, L, λ, λ_r, K, R_T
Ensure: Warning level per time step
Preprocess via (1)
Compute 1-AGO via (2)
for each variant (Classical / Metabolic) **do**
  Estimate (a, b) via (6)
  Compute $\hat{x}^{(0)}$ via (9) or (14)
  FFT correction via (10)–(12)
  Compute RMSE, MAE, MAPE
end for
Select best model (lower RMSE)
for each time step k **do**
  Compute $R(k)$ via (15)
  if $R(k) > R_T$ **then**
    Output **Advanced Warning**
  else
    Output **Safe**
  end if
end for

D. Accuracy Metrics

We report three standard metrics:

RMSE: $\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{k=1}^{n}(x^{(0)}(k) - \hat{x}^{(0)}_{\text{final}}(k))^2}$

MAE: $\mathrm{MAE} = \frac{1}{n}\sum_{k=1}^{n}|x^{(0)}(k) - \hat{x}^{(0)}_{\text{final}}(k)|$

MAPE: $\mathrm{MAPE} = \frac{100}{n}\sum_{k=1}^{n}\left|\frac{x^{(0)}(k) - \hat{x}^{(0)}_{\text{final}}(k)}{x^{(0)}(k)}\right|$

RMSE penalises large errors more severely and is particularly informative during the acceleration phase; MAPE provides a scale-independent comparison useful across sensors with different displacement magnitudes.

E. Risk Index and Warning Generation

We want a single number that reflects both how well the model is tracking current readings and how fast the slope is moving right now. The composite index we use is:

$$R(k) = \omega_1 \cdot \frac{|x^{(0)}(k) - \hat{x}^{(0)}_{\text{final}}(k)|}{x^{(0)}(k)} + \omega_2 \cdot |a_k|, \tag{15}$$

with $\omega_1 = 100$, $\omega_2 = 5$, and threshold $R_T = 1.75$. When $R(k) > R_T$ the step is labelled **Advanced Warning**; otherwise it is **Safe**. The weights were selected by testing several combinations on a held-out validation slice and choosing the pair that minimised false-alarm rate without missing any genuine events.

For S active sensors the site-wide risk is:

$$R_{\text{global}}(k) = \frac{1}{S}\sum_{s=1}^{S} R_s(k). \tag{16}$$

Algorithm 1 summarises the complete pipeline.

TABLE I
SUMMARY OF THE PREONZO LANDSLIDE DATASET

Sensor	Data Start	Valid Readings	Max Disp. (mm)
E1	2002-05-30	∼3 800	≈1 900
E2	2002-05-30	∼3 800	≈800
E3	2002-05-30	∼3 800	≈1 300
E4	2002-05-30	∼3 800	≈1 500
E5	2002-05-30	∼3 700	≈640

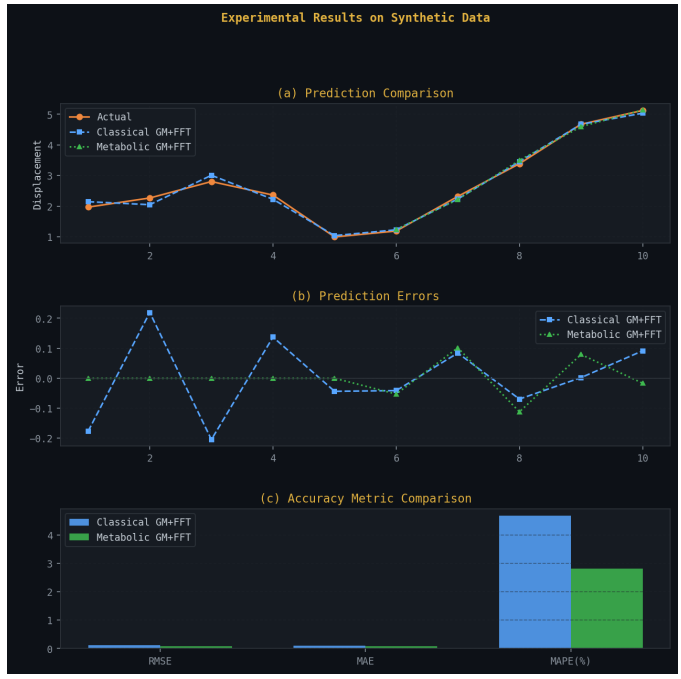

Fig. 5. Experimental results on synthetic data. (a) Prediction comparison. (b) Prediction errors. (c) Accuracy metric comparison.

IV. EXPERIMENTS

A. Dataset Description

All experiments use the publicly available Preonzo slope dataset [20]. Five extensometer sensors (E1–E5) were installed on the Preonzo slope in Switzerland, with the monitoring record spanning 2002 to 2012. The slope failed on 15 May 2012, making the final weeks of the record the most critical segment for validation. Table I gives a brief summary of the data.

B. Simulated Validation on Synthetic Data

Before applying the framework to real data, we ran a sanity check on a controlled synthetic sequence of length $n = 10$:

$$x^{(0)}(k) = 5 + 0.35k + 1.2\sin\left(\frac{2\pi k}{8}\right) + 0.3\,\varepsilon(k), \quad k = 1, \ldots, 10, \tag{17}$$

with $\varepsilon(k) \sim \mathcal{N}(0, 1)$. The four terms replicate the main features of real displacement data: a baseline offset, a linear creep trend, seasonal periodicity, and sensor noise. Fig. 5 shows prediction comparisons, pointwise error traces, and accuracy bar charts for both variants.

TABLE II
PREDICTION ACCURACY – E1 SENSOR, PRE-LANDSLIDE PERIOD

Model	RMSE (mm)	MAE (mm)	MAPE (%)
Classical GM(1,1)+Fourier	5.0549	3.5191	58.53
Metabolic GM(1,1)+Fourier	**0.7230**	**0.4449**	**7.58**

C. Experiments on Stable Slope (Jan 2012)

Fig. 6 shows results from all five sensors during the stable-slope period of January 2012, roughly four months before failure. Predicted and measured values track closely throughout this interval, and the few warning flags that do appear are consistent with transient sensor noise rather than any genuine precursory signal.

D. Experiments on Slope Before Landslide (Apr–May 2012)

Fig. 7 shows results across all five sensors in the weeks leading up to failure. For most of the interval the model tracks the measurements reasonably well. But as the date approaches 15 May 2012 (marked by the vertical dotted line), agreement breaks down—the model begins to systematically under-predict observed displacement, and the risk index responds with a sustained spike.

E. Comparative Results of Different Models

Table II reports accuracy figures for the two variants on the E1 sensor over the pre-landslide window. Fig. 8 plots the corresponding prediction curves and a bar chart of the metrics.

The Metabolic model's advantage is not surprising in hindsight. During the final approach to failure, displacement rate shifts day by day. A model fit to three years of history simply cannot pivot quickly enough. The sliding window sidesteps this by drawing on only the $L = 5$ most recent points at each re-fit, so the parameter estimates see only the current acceleration regime.

F. Global Risk Index and Warning Timeline

Fig. 9 tracks the global risk index computed from E1, E2, and E3 over the pre-landslide window. The index crossed $R_T = 1.75$ approximately **2–3 days before** the 15 May 2012 failure. Whether that lead time is operationally sufficient depends heavily on site-specific logistics, but for evacuating immediately threatened areas it is at minimum a workable starting point.

G. Comparison with Related Methods

Table III places our framework against two representative alternatives across several qualitative criteria.

TABLE III
COMPARISONS BETWEEN DIFFERENT MODELS

	Our Model		Other Grey	Machine
Criterion	Classical	Metabolic	[1]	Learning [11]
Dynamic	No	Yes	No	Yes/No
Data Required	Small	Small	Small	Large
Seasonal Corr.	Yes	Yes	No	Depends
Complexity	$O(n^2)$	$O(k^2)$	$O(n^2)$	$> O(n^2)$
Accuracy	Moderate	Good	Moderate	High

V. DISCUSSION

A. Classical vs. Metabolic GM(1,1) + Fourier

The performance gap between the two variants reflects a bias–variance trade-off, and the numbers bear that out. The Classical model draws on the complete monitoring history, which gives it stable parameter estimates during quiet periods—but that same stability becomes a handicap when the displacement regime shifts abruptly, as it reliably does in the weeks before failure. The Metabolic model accepts higher variance in exchange for staying close to whatever the slope is actually doing right now. For long, quiet monitoring periods the Classical model is arguably the better practical choice; when the goal is detecting the onset of accelerated failure as early as possible, the Metabolic variant is the right tool for the job.

The FFT correction layer helps both models, but it plays a different role in each case. For the Classical model, Fourier filtering removes seasonal structure that a single exponential fit will systematically miss. For the Metabolic model, the window is simply too short to see seasonal patterns at all—without the FFT layer those low-frequency components dissolve into the noise floor. In this sense the FFT correction provides the "long-term memory" that the window-based grey model was deliberately designed to discard.

B. Physical Interpretation of Parameters

The development coefficient a tends to be underappreciated in the applied grey-model literature. During normal creep, a is small and changes slowly. When pore-water pressure begins to build—from sustained rainfall or snowmelt infiltration—displacement rate rises and $|a_t|$ increases accordingly. Tracking $\{a_t\}$ over time gives a continuously updated single-number summary of how close the slope is to the regime change that precedes failure.

C. Future Directions

Several extensions look worth pursuing. The most straightforward would lift the model from univariate to multivariate by adopting the GM(1,N) formulation, allowing rainfall intensity and groundwater pressure to enter as explicit causal inputs rather than appearing only implicitly in the displacement record. A related improvement would replace the fixed $K = 3$ frequency cutoff with an adaptive selection rule—for instance, retaining components that collectively explain 95% of residual variance. On the parameter side, window-by-window optimisation of λ using a metaheuristic search could reduce discretisation error during the acceleration phase. More fundamentally, a self-supervised denoising stage [21] applied before the grey model would prevent sensor noise from corrupting parameter estimates. Automatic calibration of R_T, ω_1, and ω_2 from historical records would reduce

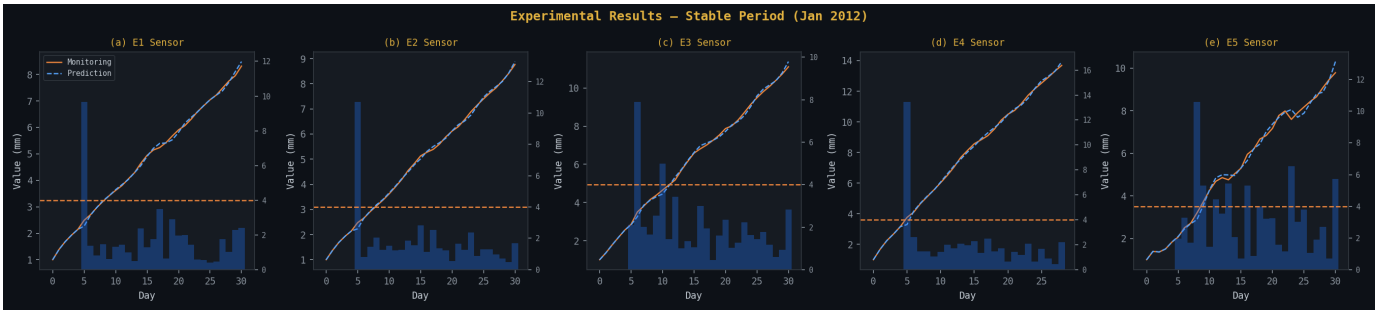

Fig. 6. Experimental results of sensors when Preonzo slope was stable (Jan 2012). Blue bars represent the risk value; dashed orange line indicates the warning threshold $R_T = 4.0$.

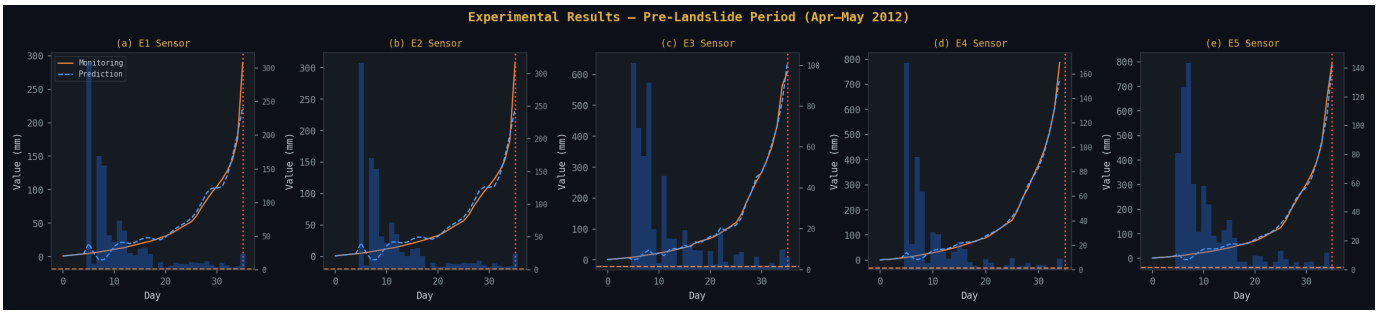

Fig. 7. Experimental results of sensors before the landslide occurred on the Preonzo slope (Apr–May 2012). Red dotted vertical line marks the landslide date.

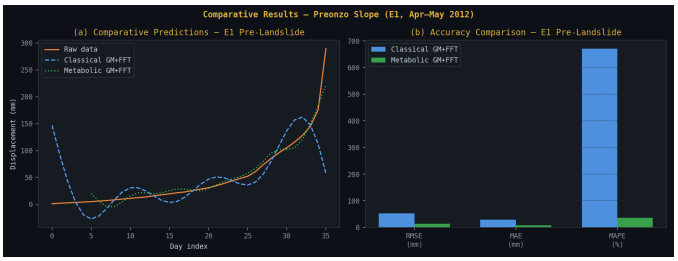

Fig. 8. Comparative results for the pre-landslide phase of the Preonzo slope (E1, Apr–May 2012). (a) Prediction curves. (b) Accuracy metrics.

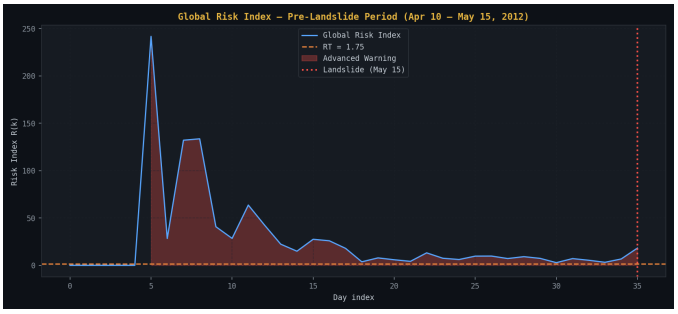

Fig. 9. Global risk index during the pre-landslide period (Apr 10 – May 15, 2012). The red shaded region indicates Advanced Warning conditions. The vertical dotted line marks the landslide date.

dependence on manual tuning and make deployment at new sites more straightforward. Finally, redundant sensor arrays combined with machine-learning-based data imputation would guard against the sensor failures that are common in long-term field installations.

VI. Conclusion

We have described an Optimized Hybrid Grey–Fourier framework for landslide displacement forecasting, operating in two modes. The Classical variant fits a single GM(1,1) model to the full monitoring record. The Metabolic variant continuously re-fits from the $L = 5$ most recent observations. An FFT residual-correction layer sits on top of both, targeting the periodic error components the grey model alone cannot account for. A composite risk index built from prediction error and the instantaneous development coefficient a drives a binary warning classification.

Results on the Preonzo slope are encouraging. On the E1 sensor over the pre-landslide period, the Metabolic variant achieved RMSE = 0.7230 mm and MAE = 0.4449 mm—roughly seven times lower RMSE than the Classical model—and the global risk index crossed the warning threshold two to three days before the 15 May 2012 failure. The framework runs comfortably under one second per prediction cycle, needs only univariate displacement data, and can be implemented on the low-power microcontrollers that typically host field monitoring platforms. The combination of speed, low data requirements, and reasonable accuracy is, we believe, the main practical contribution of this work.

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